

A primer on structured decision making for invasive species (using dynamic programming)

Olivier Gimenez^{1*} Abby Keller^{2*} Lucile Marescot^{3*} Cassie Speakman^{4*}

2025-12-31

¹ CEFE, Univ Montpellier, CNRS, EPHE, IRD, Montpellier, France ² address ³ address ⁴ address

* Corresponding author: olivier.gimenez@cefe.cnrs.fr

Invasive species (IAS) pose significant ecological, economic, and public health challenges.

Robust methods are needed for guiding effective management. Overall, we hope to demonstrate

the flexibility and utility of structured decision making in invasive species management.

Target journal: Biological invasions? NeoBiota?.

Keywords: Adaptive management, Invasive species, Stochastic dynamic programming, Structured decision making

Introduction

Invasive species are a significant global issue, with wide-ranging impacts on ecosystems, economies, and public health (Pyšek et al. 2020, Roy et al. 2024).

Effective management of invasive species requires blabla for guiding control efforts and evaluating the success of eradication or regulation programs (Williams et al. 2002, Thompson et al. 2021).

Pitch: Structured decision-making (SDM) and adaptive management (AM) have been repeatedly recommended for the management of invasive species, yet they remain underused in practice. There are many reasons for this gap, one of which is the technical challenge of formally identifying and quantifying trade-offs among management objectives. Here, we contribute to ongoing efforts by illustrating the use of stochastic dynamic programming (SDP) and some of its variants — including active and passive adaptive management and partially observable Markov decision processes (POMDPs) — across different ecological contexts (a single population, a metapopulation, and two interacting populations).

Several papers have recommended or implemented structured decision-making (SDM) for managing invasive species. It's worth providing a short review of this literature.

In a recent review on impacts, the use of ML to guide management actions identified as a future research direction (see Appendix 3 in Haubrock et al. (2025)).

“Another recent research direction is the use of machine learning to aid the development of conservation management strategies. These methods involve models that integrate geospatial, climate, species distribution and other data to simulate the distribution of species and vegetation types under different land use and conservation planning scenarios. Reinforcement Learning algorithms can then be used to learn a management policy that attains a specified quantifiable objective (such as minimising the number of individuals from a non-native species) under

constraints such as cost or the amount of land that can be protected. Dietterich, Taleghan and Crowley (2013) demonstrated how Reinforcement Learning can be applied to managing a river system to control the expansion of a non-native plant species. Lapeyrolerie et al. (2022) demonstrated the potential of Reinforcement Learning for resource management problems, such as setting fishing quotas. Conservation area prioritisation through artificial intelligence (CAPTAIN) (Silvestro et al., 2022) is a general simulation framework that allows the user to define flexibly management objectives to be optimised. The unprecedented rate at which new technologies are emerging holds great promise for improving impact assessments related to biological invasions.”

Coypu (Myocastor coypu) as a case study throughout the paper. Code and tutorials available as appendices.

Add a shiny for each case study so that folks can play around?

A primer on structured decision making

Why and what (all)

Stochastic dynamic programming (Olivier)

Example on computer maintenance (coverted into coypu example) to explain SDP on a simple example ?

Case studies with coypus

Single population (Olivier)

Take wolf example and turn it into a coypu example ?

Two interacting species (Lucile)

Competition or predator-prey, or host-pathogen / disease thing. See SDP-species-interactions/
in Olivier biblio repo.

Metapopulation (Olivier)

Accounting for uncertainty

POMDP (Cassie)

Illustrate POMDP, Olivier has some code extending the computer maintenance example from
above.

<https://besjournals.onlinelibrary.wiley.com/doi/10.1111/2041-210X.14374>

<https://besjournals.onlinelibrary.wiley.com/doi/10.1111/2041-210X.13692>

Adaptive management (Abby)

When is it useful. Start with value of information?

On a single species ; check out Conroy's book, there is a nice figure illustrating AM.

<https://onlinelibrary.wiley.com/doi/10.1002/ece3.71176>

Discussion (all)

Main messages

How is SDM relevant.

Limitations

Curse of dimensionality.

Perspectives

Acknowledgments

Data availability statement

Data and code are available at <https://github.com/oliviergimenez/SDPaper>.

References

- Haubrock, P. J., T. Everts, N. A. S. Abreo, J. Bojko, V. Deklerck, J. W. E. Dickey, A. C. S. Franco, E. García-Berthou, S. Katsanevakis, N. I. Kirichenko, S. Mammola, M. A. Nuñez, B. Parker, R. Scalera, I. Soto, D. Strubbe, A. S. Tarkan, L. Vilizzi, T. Adriaens, P. Balzani, D. Blońska, E. Briski, R. Brys, A. L. Burgess, J. E. Byers, C. Cano-Barbacil, G. Castaldelli, J. T. A. Dick, V. Dominguez Almela, R. D. Dimarco, M. Florencio, A. Kouba, M. Kourantidou, I. Kurtul, I. Martín-Forés, O. Morissette, J. D. Olden, B. E. Soares, J. Truszkowski, H. Verreycken, M. Kenis, R. Sousa, and J. R. Britton. 2025. The impacts of biological invasions. *Biological Reviews* n/a.
- Pyšek, P., P. E. Hulme, D. Simberloff, S. Bacher, T. M. Blackburn, J. T. Carlton, W. Dawson, F. Essl, L. C. Foxcroft, P. Genovesi, J. M. Jeschke, I. Kühn, A. M. Liebhold, N. E. Mandrak, L. A. Meyerson, A. Pauchard, J. Pergl, H. E. Roy, H. Seebens, M. van Kleunen, M. Vilà, M. J. Wingfield, and D. M. Richardson. 2020. Scientists' warning on invasive alien species. *Biological Reviews* 95:1511–1534.

Roy, H. E., A. Pauchard, P. J. Stoett, T. Renard Truong, L. A. Meyerson, S. Bacher, B. S. Galil, P. E. Hulme, T. Ikeda, S. Kavileveettil, M. A. McGeoch, M. A. Nuñez, A. Ordonez, S. J. Rahlao, E. Schwindt, H. Seebens, A. W. Sheppard, V. Vandvik, A. Aleksanyan, M. Ansong, T. August, R. Blanchard, E. Brugnoli, J. K. Bukombe, B. Bwalya, C. Byun, M. Camacho-Cervantes, P. Cassey, M. L. Castillo, F. Courchamp, K. Dehnen-Schmutz, R. D. Zenni, C. Egawa, F. Essl, G. Fayvush, R. D. Fernandez, M. Fernandez, L. C. Foxcroft, P. Genovesi, Q. J. Groom, A. I. González, A. Helm, I. Herrera, A. J. Hiremath, P. L. Howard, C. Hui, M. Ikegami, E. Keskin, A. Koyama, S. Ksenofontov, B. Lenzner, T. Lipinskaya, J. L. Lockwood, D. C. Mangwa, A. F. Martinou, S. M. McDermott, C. L. Morales, J. Müllerová, N. A. Mungi, L. K. Munishi, H. Ojaveer, S. N. Pagad, N. P. K. T. S. Pallewatta, L. R. Peacock, E. Per, J. Pergl, C. Preda, P. Pyšek, R. K. Rai, A. Ricciardi, D. M. Richardson, S. Riley, B. J. Rono, E. Ryan-Colton, H. Saeedi, B. B. Shrestha, D. Simberloff, A. Tawake, E. Tricarico, S. Vanderhoeven, J. Vicente, M. Vilà, W. Wanzala, V. Werenkraut, O. L. F. Weyl, J. R. U. Wilson, R. O. Xavier, and S. R. Ziller. 2024. Curbing the major and growing threats from invasive alien species is urgent and achievable. *Nature Ecology & Evolution* 8:1216–1223.

Thompson, B. K., J. D. Olden, and S. J. Converse. 2021. Mechanistic invasive species management models and their application in conservation. *Conservation Science and Practice* 3:e533.

Williams, B. K., J. D. Nichols, and M. J. Conroy. 2002. Analysis and management of animal populations. Academic Press, San Diego, California, USA.